Autonomous control: predictions, deadlines and fallbacks

Model predictive control, or MPC, is a natural place to ask where quantum optimization could fit. At each step, the controller uses a model to predict how the system will evolve over a short horizon, solves for the best sequence of inputs, applies only the first one, then measures the new state and solves again. The prediction supplies foresight. The constant re-solving supplies feedback, so errors in the model are corrected before they compound.

That re-solving runs against a hard deadline. If an MPC solve for a fast-moving drone takes longer than the control period, the answer arrives after the moment it was meant for. Today's QPU access involves queues, network round trips, repeated sampling and classical post-processing, which makes it a poor fit for the innermost loop. One distinction is worth keeping straight here: this discussion concerns quantum optimization used for a robot, not the classical control engineering used to operate a quantum computer, which is a separate field.

A sensible arrangement keeps the fast loop local and classical, and treats any quantum solver as an optional advisor on slower timescales: re-sequencing a route, reshuffling a task list or refreshing a library of candidate motions. Its proposals pass a classical feasibility check before anything reaches the actuators. If a quantum result arrives late, breaks a constraint or loses to a classical answer, the system uses the classical one, and safety monitors, emergency stops and certified limits stay classical throughout. Figure 2 illustrates this as a conceptual pattern, not a description of any deployed system.

Trends worth watching

A few developments look genuinely important, each with a boundary. Error correction is becoming the main yardstick for hardware progress, shifting attention from how many qubits a chip holds towards how long logical information survives and how many physical qubits each logical one costs. Robotics workloads would need deep, reliable circuits, so this is the measure to follow.

Hybrid workflows are the norm rather than the exception. QAOA already leans on a classical optimizer, and practical pipelines decompose a large problem classically, send a bounded subproblem to a QPU, and stitch the results back together. Much of the interesting engineering lives in that decomposition step.

On the robotics side, embodied models and simulation are moving quickly. NVIDIA, for example, describes world foundation models, synthetic-data pipelines and multimodal robot policies trained on a mix of real and synthetic experience. These are classical tools, and simulated physics never matches reality perfectly, but they create digital-twin environments where alternative plans, including those from experimental solvers, can be compared safely before they touch hardware.

Benchmarking is maturing as well. Reviews of quantum optimization emphasise application-level comparisons against strong classical methods, judged on solution quality, time to solution and overall cost, rather than raw qubit counts. For anyone weighing a robotics claim, that is the right lens.

Use cases: tested, studied and imagined

Sorting the evidence by maturity helps. The table below separates what has been tried on a real industrial problem from what remains a research proposal or an illustrative idea.

Use case	Status	What the evidence supports
Sequencing robot paths in an automotive work cell (BMW sealant application)	Tested on a real industrial problem	Hybrid quantum and classical methods compared; a classical genetic-algorithm pipeline performed best, and annealer mapping overhead blocked industrial scale
Robot arm posture (inverse kinematics)	Research proposal	Quantum formulation proposed; no hardware or practical speedup claimed
Nonlinear MPC recast as a QUBO	Research proposal	Benefit explicitly conditional on an efficient quantum computer
Warehouse fleet routing, task assignment and charging schedules	Illustrative candidate	Combinatorial structure suits QUBO encoding; no deployment demonstrated
Drone or inspection mission allocation	Potential	Slow mission-level assignment only; flight control and collision avoidance remain local and classical
Offline motion libraries and digital-twin plan comparison	Proposed pattern	Candidates from any solver validated in simulation before use
Materials for batteries or grippers	Prospective, longer term	Indirect route: quantum simulation of chemistry might someday inform hardware design